

Annex 179B

(normative)

Test fixtures for 200GBASE-CR1, 400GBASE-CR2, 800GBASE-CR4, 1.6TBASE-CR8, 200GAUI-1 C2M, 400GAUI-2 C2M, 800GAUI-4 C2M, and 1.6TAUI-8 C2M

179B.1 Test fixtures

Transmitter and receiver measurements at TP2 or TP3 for the 200GBASE-CR1, 400GBASE-CR2, 800GBASE-CR4, and 1.6TBASE-CR8 hosts (see 179.8.1) and at TP1a or TP4a (see Figure 176D–4) for the 200GAUI-1, 400GAUI-2, 800GAUI-4, and 1.6TAUI-8 C2M hosts (see Annex 176D), are made utilizing the test fixture specified in 179B.2.

Cable assembly measurements for the cable assembly types (see Annex 179D) are made between TP1 and TP4 with test fixtures as specified in 179B.3 on both ends. The test fixtures are specified in a mated state to enable connections to measurement equipment. The reference insertion loss of the mated test fixtures is 9.75 dB at 53.125 GHz using Equation (179B–5).

Module measurements for modules specified in Annex 176D are made at module compliance points TP1 and TP4 (see Figure 176D–5) with test fixtures as specified in 179B.3.

The requirements in the referenced subclauses are not the MDI connector specifications for an implemented design.

179B.1.1 Reference Impedance

The reference impedance for differential specifications is 92.5 Ω . The reference impedance for common-mode specifications is 23.125 Ω .

NOTE—Calculation of modal S-parameters may require transformation of the measured single-ended S-parameters to the specified reference impedance. See 178A.1.3.

179B.2 TP2 or TP3 test fixture

The TP2 or TP3 test fixture (also known as Host Compliance Board) is required for measuring the transmitter and receiver specifications at TP2 and TP3. The test fixture has an edge connector interface that is compatible with the appropriate MDI connector on the Host board, and provides a high-speed electrical path between the MDI connector and the coaxial connector that defines the TP2 or TP3 test point. The TP2 and TP3 test points are illustrated in Figure 179A–1.

The TP2 or TP3 test fixture is also used in AUI-C2M testing for Host input and output compliance. The TP2 and TP3 test points correspond to the Annex 176D test points TP1a and TP4a, respectively. The TP1a and TP4a test points are illustrated in Figure 176D–4.

179B.2.1 TP2 or TP3 test fixture insertion loss

The TP2 or TP3 test fixture reference insertion loss is defined as the insertion loss between the reference plane of the coaxial connector and the mating point of the MDI connector. The reference insertion loss is

defined by Equation (179B–1) and illustrated by Figure 179B–1. The effects of differences between the insertion loss of an actual test fixture and the reference insertion loss are to be accounted for in the measurements.

$$ILdd_{\text{tfref}}(f) = 0.3073\sqrt{f} + 0.01172f + 0.0003321f^2 \quad (179B-1)$$

where

$ILdd_{\text{tfref}}(f)$ is the test fixture reference insertion loss in dB at frequency f
 f is the frequency in GHz

179B.3 Cable assembly test fixture

The cable assembly test fixture (also known as Module Compliance Board) is required for measuring the cable assembly specifications in 179.11 at TP1 and TP4. The test fixture has an MDI connector, and provides a high-speed electrical path between the MDI connector and the coaxial connector that defines the TP1 or TP4 test point. The TP1 and TP4 test points are illustrated in Figure 179A–1.

The cable assembly test fixture is also used in AUI-C2M testing for module input and output compliance. The TP1 and TP4 test points for AUI-C2M testing are illustrated in Figure 176D–5.

179B.3.1 Cable assembly test fixture insertion loss

The cable assembly test fixture reference insertion loss is defined as the insertion loss between the reference plane of the coaxial connector and the mating point of the MDI connector. The reference insertion loss is defined by Equation (179B–2) and illustrated by Figure 179B–1. The specific MDI connector configuration used to define the reference insertion loss is the “Single Row Surface Mount Technology Connector and Cage” defined in the OSFP1600 specification (see 179C.2.3). Differences between the insertion loss of an actual test fixture and the reference insertion loss, including those arising from other MDI connectors, shall be accounted for in the measurements. Usage of the the Single Row Surface Mount Technology Connector and Cage for cable assembly test fixtures is recommended.

The insertion loss of the actual test fixture is equal to the measured loss of the actual test fixture with a TP2 or TP3 test fixture minus the loss of the specific TP2 or TP3 test fixture used in that measurement.

$$ILdd_{\text{catfref}}(f) = 0.0912 + 0.3102\sqrt{f} + 0.008578f + 0.000759f^2 \quad (179B-2)$$

where

$ILdd_{\text{catfref}}(f)$ is the test fixture reference insertion loss in dB at frequency f
 f is the frequency in GHz

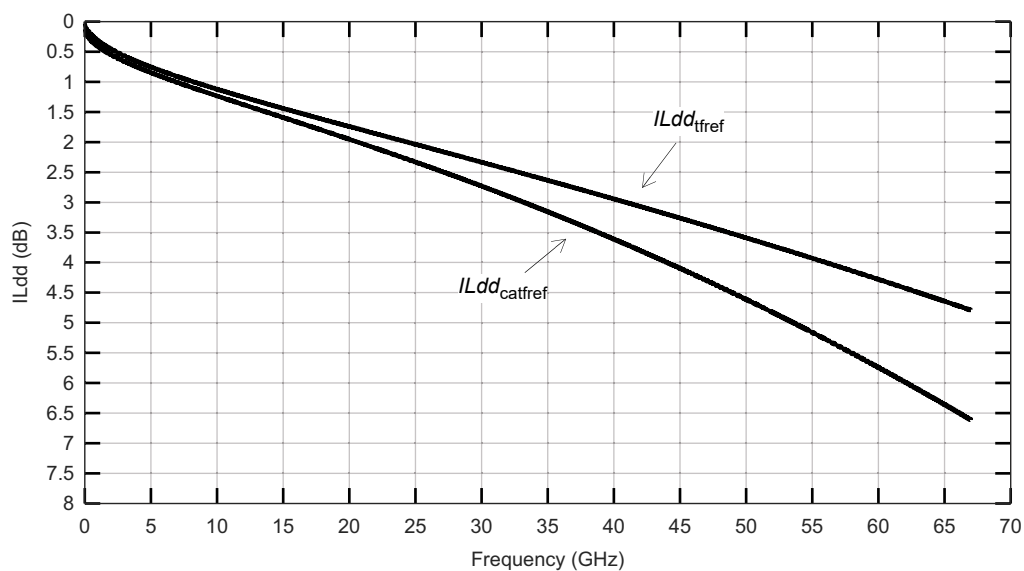


Figure 179B-1—Test fixtures reference insertion losses

179B.4 Mated test fixtures

The TP2 or TP3 test fixture and the cable assembly test fixture are specified in a mated state illustrated in Figure 92–18. The mated test fixtures specifications are given below.

179B.4.1 Mated test fixtures insertion loss

The insertion loss of the mated test fixtures shall meet Equation (179B–3) and Equation (179B–4).

$$ILdd(f) \leq ILdd_{MTFmax}(f) = 0.1 + 0.8\sqrt{f} + 0.0055f + 0.00125f^2 \quad (179B-3)$$

$$ILdd(f) \geq ILdd_{MTFmin}(f) = 0.3854\sqrt{f} + 0.0455f + 0.00045f^2 \quad (179B-4)$$

for $0.01 \leq f \leq 67$

where

$ILdd(f)$	is the measured insertion loss in dB at frequency f
$ILdd_{MTFmax}(f)$	is the maximum mated test fixtures insertion loss in dB at frequency f
$ILdd_{MTFmin}(f)$	is the minimum mated test fixtures insertion loss in dB at frequency f
f	is the frequency in GHz

The reference insertion loss of the mated test fixtures is determined using Equation (179B–5).

$$ILdd_{MTFref}(f) = 0.0912 + 0.6175\sqrt{f} + 0.0203f + 0.001091f^2 \quad (179B-5)$$

where

$ILdd_{MTFref}(f)$	is the reference insertion loss of the mated test fixtures in dB at frequency f
f	is the frequency in GHz

NOTE—The reference insertion loss of the mated test fixtures is equal to the sum of Equation (179B–1) and Equation (179B–2).

The mated test fixtures insertion loss limits are illustrated in Figure 179B–2.

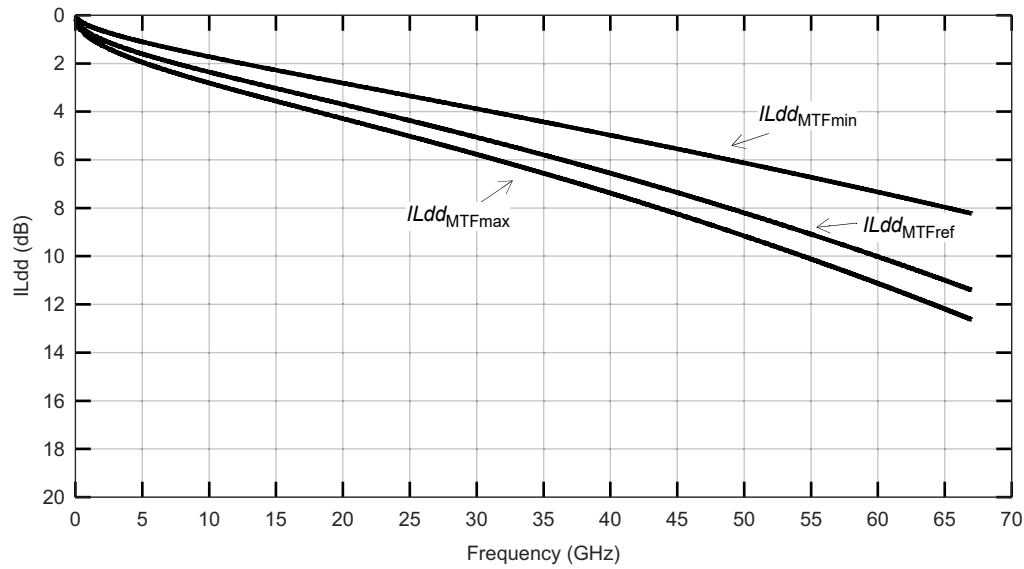


Figure 179B-2—Mated test fixtures insertion loss

The FOM_{ILD} is calculated according to 93A.4 with $f_b = 106.25$ GHz, $T_i = 5$ ps, and $f_r = 0.55 \times f_b$. The fitted insertion loss and insertion loss deviation are computed over the range $f_{min} = 0.05$ GHz to $f_{max} = 67$ GHz, with $\Delta f = 0.01$ GHz. FOM_{ILD} shall be less than or equal to 0.15 dB.

179B.4.2 Mated test fixtures return loss

179B.4.2.1 Mated test fixtures effective return loss (ERL)

The values of the mated test fixtures ERL are computed using the procedure in 93A.5 with the parameter values in Table 179B–1. Parameters that do not appear in Table 179B–1 take values from Table 179–24.

The reference differential impedance for the mated test fixtures ERL computation shall be 92.5 Ω . The reference differential impedance for the mated test fixtures ERL computation is independent of the reference differential impedance for the s-parameter data, 179B.1.1. The mated test fixtures ERL shall be greater than or equal to 10.3 dB, with T_{fx} equal to 0.15ns. T_{fx} is an allowance for discontinuities associated with the coaxial connector, and is fixed for all test fixture implementations. The mated test fixtures ERL shall be greater than or equal to 9 dB, with T_{fx} equal to 0ns.

Table 179B–1—Mated test fixtures ERL parameter values

Parameter	Symbol	Value	Units
Transition time associated with a pulse	T_r	0.005	ns
Incremental available signal loss factor	β_x	0	GHz
Permitted reflection from a transmission line external to the device under test	ρ_x	0.618	—
Length of the reflection signal	N	1600	UI
Equalizer length associated with reflection signal	N_{bx}	0	UI
Tukey window flag	tw	1	—
Number of samples per unit interval	M	32	—
Target detector error ratio	DER_0	2×10^{-5}	—
NOTE—The ERL is computed twice, with different values for T_{fx} , to ensure the quality of the coaxial connectors used in the test fixtures.			

179B.4.2.2 Mated test fixtures modal ERL

Mated test fixtures modal ERL (ERL_{CC} or ERL_{DC}) is the ERL computed according to 93A.5 using the parameter values in Table 179B–1 and Table 179–24 and T_{fx} equal to 0.15ns, with the following exceptions:

- For mated test fixtures TP1 ERL_{CC} , $s_{ii}(f)$ is replaced with the MTF $s_{CC11}(f)$.
- For mated test fixtures TP1 ERL_{DC} , $s_{ii}(f)$ is replaced with the MTF $s_{DC11}(f)$.
- For mated test fixtures TP2 ERL_{CC} , $s_{ii}(f)$ is replaced with the MTF $s_{CC22}(f)$.
- For mated test fixtures TP2 ERL_{DC} , $s_{ii}(f)$ is replaced with the MTF $s_{DC22}(f)$.

179B.4.2.3 Mated test fixtures modal return loss specifications

The mated test fixtures modal return loss are common-mode to common-mode and common-mode to differential-mode.

The mated test fixtures common-mode to common-mode return loss specification is satisfied by either meeting the requirement of Equation (179B–6) or having ERL_{CC} higher than the value in Table 179B–2.

The mated test fixtures common-mode to differential-mode return loss specification is satisfied by either meeting the requirement of Equation (179B–7) or having ERL_{DC} higher than the value in Table 179B–2.

Table 179B–2— Mated test fixtures modal return loss specifications

Specification	RL mask equation	RL mask figure	Modal ERL type	Modal ERL minimum (dB)
Common-mode to Common-mode	Equation (179B–6)	Figure 179B–3	ERL_{CC}	2
Common-mode to Differential-mode	Equation (179B–7)	Figure 179B–4	ERL_{DC}	17

$$RL_{cc}(f) \geq \begin{cases} 3 & 0.05 \leq f < 4 \\ 3 + \frac{2}{40}(f - 4) & 4 \leq f < 44 \\ 5 + \frac{2}{16}(44 - f) & 44 \leq f < 60 \\ 3 & 60 \leq f \leq 67 \end{cases} \quad (179B-6)$$

where

$RL_{cc}(f)$ is the common-mode to common-mode return loss in dB at frequency f
 f is the frequency in GHz

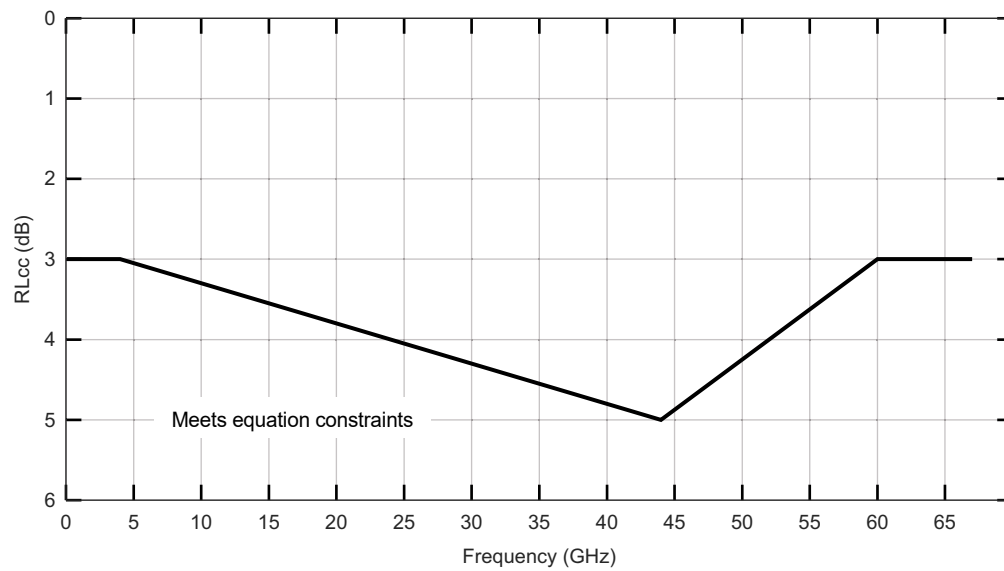


Figure 179B-3—Mated test fixtures common-mode to common-mode return loss requirements

$$RL_{dc}(f) \geq \begin{cases} 26 - 22 \frac{f}{106.25} & 0.05 \leq f < 53.125 \\ 15 & 53.125 \leq f < 67 \end{cases} \tag{179B-7}$$

where
 $RL_{dc}(f)$ is the common-mode to differential-mode return loss in dB at frequency f
 f is the frequency in GHz

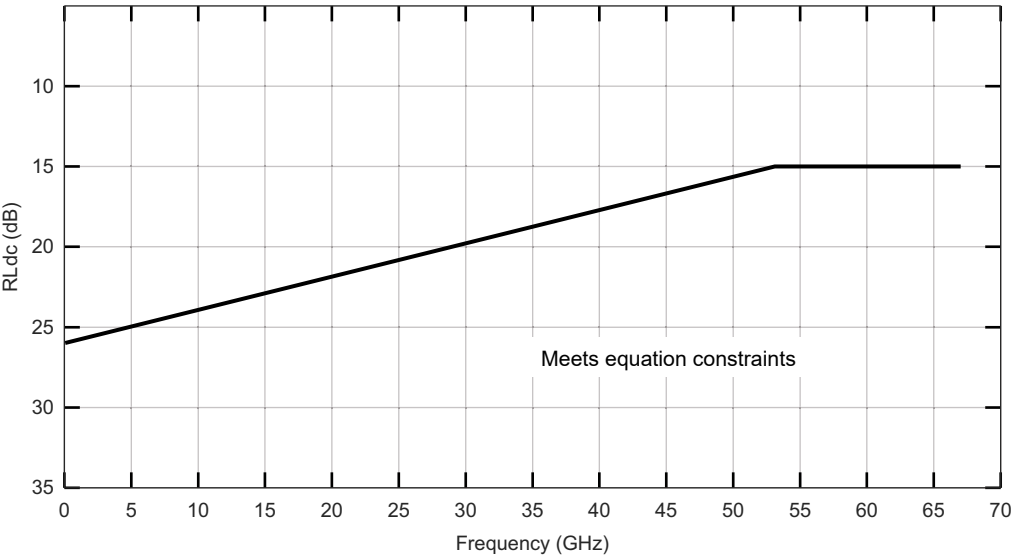


Figure 179B-4—Mated test fixtures common-mode to differential-mode return loss requirements

179B.4.3 Mated test fixtures common-mode to differential-mode insertion loss

The mated test fixtures shall comply with the following requirements:

- 1) The channel signal to common-mode ratio ($SCMR_{CH}$) is greater than 12 dB using the procedure described in 179.11.5.
- 2) The common-mode to differential-mode insertion loss of the mated test fixtures measured in both directions meets Equation (179B–8) as illustrated in Figure 179B–5.

$$IL_{dc}(f) \geq \begin{cases} 30 - 3\frac{f}{7} & 0.01 \leq f < 35 \\ 15 & 35 \leq f < 67 \end{cases} \quad (179B-8)$$

where

$IL_{dc}(f)$ is the common-mode to differential-mode insertion loss in dB at frequency f
 f is the frequency in GHz

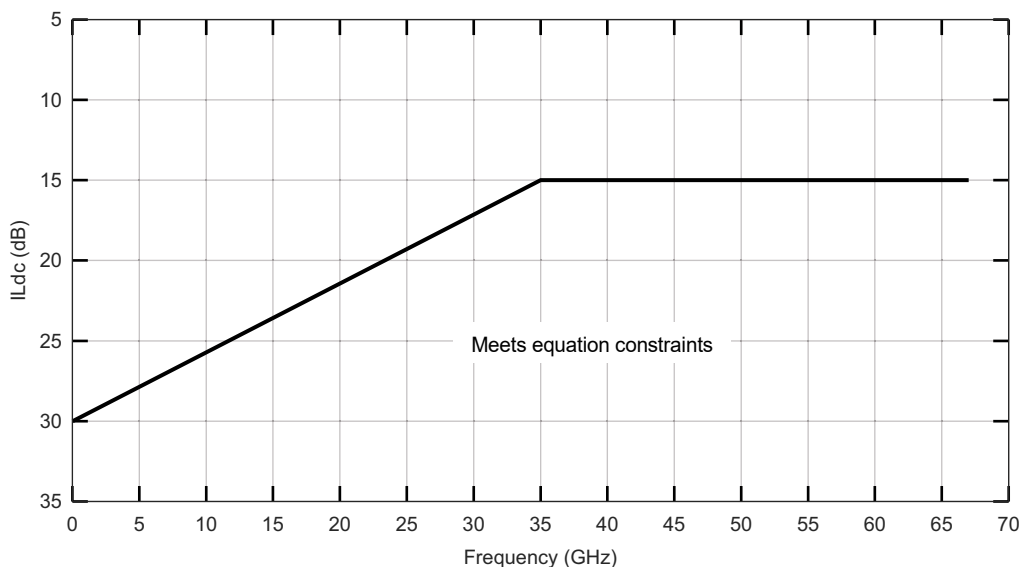


Figure 179B–5—Mated test fixtures common-mode to differential-mode insertion loss requirements

179B.4.4 Mated test fixtures integrated crosstalk noise

The mated test fixtures integrated crosstalk noise voltage is determined from the appropriate disturber loss measured over N uniformly spaced frequencies f_n spanning the frequency range 0.05 GHz to 67 GHz with maximum frequency spacing of 10 MHz, and the parameters shown in Table 179B–3.

Table 179B–3—Mated test fixtures integrated crosstalk noise parameters

Description	Symbol	Value	Units
Symbol rate	f_b	106.25	GBd
3 dB reference receiver bandwidth	f_r	58.4375	GHz
Near-end disturber peak differential output amplitude	A_{nt}	600	mV
Far-end disturber peak differential output amplitude	A_{ft}	600	mV
Near-end disturber 20% to 80% rise and fall times	T_{nt}	4.25	ps
Far-end disturber 20% to 80% rise and fall times	T_{ft}	4.25	ps

The mated test fixtures integrated crosstalk noise voltages are determined using Equation (92–44) through Equation (92–48), given the multiple disturber near-end crosstalk loss $MDNEXT_{loss}(f)$ and multiple disturber far-end crosstalk loss $MDFEXT_{loss}(f)$. The mated test fixtures integrated crosstalk noise shall meet the specifications in Table 179B–4.

Table 179B–4—Mated test fixtures integrated crosstalk noise voltage

Parameters	Value	Units
MDFEXT integrated crosstalk noise voltage (max)	4.2	mV
MDNEXT integrated crosstalk noise voltage (max)	1.5	mV

NOTE—MDFEXT is only applicable to MDI connectors with more than one lane. The number of aggressors in an MDFEXT or MDNEXT result is determined by the MDI connector type.

179B.5 Protocol implementation conformance statement (PICS) proforma for Annex 179B, Test fixtures for 200GBASE-CR1, 400GBASE-CR2, 800GBASE-CR4, 1.6TBASE-CR8, 200GAUI-1 C2M, 400GAUI-2 C2M, 800GAUI-4, and 1.6TAUI-8 C2M⁹

179B.5.1 Introduction

The supplier of a protocol implementation that is claimed to conform to Annex 179B, Test fixtures for 200GBASE-CR1, 400GBASE-CR2, 800GBASE-CR4, 1.6TBASE-CR8, 200GAUI-1 C2M, 400GAUI-2 C2M, 800GAUI-4 C2N and 1.6TAUI-8 C2M, shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in 21.

179B.5.2 Identification

179B.5.2.1 Implementation identification

Supplier ¹	
Contact point for inquiries about the PICS ¹	
Implementation Name(s) and Version(s) ^{1, 3}	
Other information necessary for full identification—e.g., name(s) and version(s) for machines and/or operating systems; System Name(s) ²	
NOTE 1—Required for all implementations. NOTE 2—May be completed as appropriate in meeting the requirements for the identification. NOTE 3—The terms Name and Version should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model).	

179B.5.2.2 Protocol summary

Identification of protocol standard	IEEE Std 802.3dj-202x, Annex 179B, Test fixtures for 200GBASE-CR1, 400GBASE-CR2, 800GBASE-CR4, 1.6TBASE-CR8, 200GAUI-1 C2M, 400GAUI-2 C2M, 800GAUI-4 C2M, and 1.6TAUI-8 C2M
Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS	
Have any Exception items been required? No [] Yes [] (See 21; the answer Yes means that the implementation does not conform to IEEE Std 802.3dj-202x.)	

Date of Statement	
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⁹Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this annex so that it can be used for its intended purpose and may further publish the completed PICS.

179B.5.3 Major capabilities/options

Item	Feature	Subclause	Value/Comment	Status	Support
HTF1	TP2 or TP3 test fixture	179B.2		O	Yes [] No []
CATF1	Cable assembly test fixture	179B.3		O	Yes [] No []

179B.5.4 PICS proforma tables for test fixtures for Test fixtures for 200GBASE-CR1, 400GBASE-CR2, 800GBASE-CR4, 1.6TBASE-CR8, 200GAUI-1 C2M, 400GAUI-2 C2M, 800GAUI-4 C2M and 1.6TAUI-8 C2M

Item	Feature	Subclause	Value/Comment	Status	Support
TF1	Mated test fixtures insertion loss	179B.4.1	Per Equation (179B–3) and Equation (179B–4)	M	Yes []
TF2	Mated test fixtures FOM_{ILD}	179B.4.1	Less than or equal to 0.15 dB	M	Yes []
TF3	Mated test fixtures ERL	179B.4.2	Greater than or equal to 10.3 dB	M	Yes []
TF4	Mated test fixtures common-mode to differential-mode insertion loss	179B.4.3		M	Yes []
TF5	Mated test fixtures common-mode to common-mode return loss	179B.4.2.3	Per Table (179B–2)	M	Yes []
TF6	Mated test fixtures common-mode to differential-mode return loss	179B.4.2.3	Per Table (179B–2)	M	Yes []
TF7	Mated test fixtures integrated crosstalk noise	179B.4.4	Meets the requirements of Table 179B–4	M	Yes []